

OSAMA ALI NAJI

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PROFILE

AI Engineer with First-Class Honors in Computer Science and hands-on experience building and deploying production AI systems. Specialize in ML & DL, Agentic AI, RAG pipelines, LLM integration, and Computer Vision — delivering end-to-end solutions using LangChain, n8n, PyTorch, and YOLO. Proven ability to work independently, having designed ML & DL models, AI automation agents, and deep learning systems at Tytan KSA. Actively building multi-agent systems and real-world AI products. Dedicated to deploying scalable AI solutions that drive operational efficiency and business growth.

EDUCATION

Taibah University

Jan 2020 – May 2024

Bachelor of Science in Computer Science

- GPA: **4.95/5.00** - First Class Honors.
- Graduation Project: Phishing Attack Detection Using AI. (A+)
- Coursework: Data Structures & Algorithms, Operating Systems, Networks, Software Engineering.

KAUST (King Abdullah University of Science and Technology)

Oct 2024 – Feb 2025

Specialization — Artificial Intelligence

- Focused study in AI concepts, tools, and practical applications.

Apple Developer Academy | TUWAIQ

Oct 2025 – Nov 2025

Education Scholarship — iOS Dev, Swift, UI/UX

- Scholarship program in app development, Swift coding, and UI/UX design fundamentals.

WORK EXPERIENCE

AI Engineer

Aug 2025 – Present

Tytan KSA | Full-time | On-site

- Independently led the AI division — designed, built, and deployed ML/DL models for delivery performance optimization with no team or senior guidance.
- Built ML models for delivery time prediction and route optimization, improving operational decision-making.
- Developed Power BI dashboards that gave management real-time visibility into delivery KPIs and bottlenecks.
- Built AI automation agents to streamline internal workflows and reduce manual operations.

AI, Backend & Database Intern

Mar 2025 – Jun 2025

SIT – IT Solutions | Internship

- Gained hands-on experience in database engineering, data science, AI integration, and backend development.
- Built AI-integrated applications, designed relational databases, and developed RESTful APIs.

Research & Development Leader

Nov 2024 – Jun 2025

Wae (وعي) | Volunteer | Riyadh, Hybrid

- Led AI awareness workshops targeting diverse community audiences across Saudi Arabia.
- Directed AI-driven R&D initiatives and developed educational content on emerging AI technologies.

Artificial Intelligence Intern

Jul 2023 – Aug 2023

Taibah Valley | Internship | Medina

- Developed a high-accuracy Arabic sentiment analysis model using Python, NLTK, and scikit-learn.
- Conducted data acquisition, cleaning, and preprocessing pipelines for NLP model training.

PROJECTS

Hotel Customer Service Agent — WhatsApp AI Bot | *LangGraph · LangChain · RAG · EvolutionAPI · OpenAI API · Chroma DB* [LINK](#)

- **Built an end-to-end Agentic AI system** that handles hotel customer inquiries automatically via WhatsApp — room availability, pricing, booking FAQs, and escalation to human staff.
- **Implemented RAG pipeline** connecting hotel knowledge base (policies, room types, services) to an LLM, enabling accurate and context-aware responses in Arabic and English.
- **Orchestrated the full workflow** — integrating WhatsApp API, vector database retrieval, LLM response generation, and fallback logic in a single automated pipeline.

Company Document Chatbot — RAG-Powered AI | *LangChain · FastAPI · Qdrant · Cohere API* [LINK](#)

- **Built an end-to-end RAG chatbot** that answers employee questions based exclusively on uploaded company documents (PDF, DOCX, TXT) eliminating hallucinations by grounding every response in retrieved document chunks with source attribution.
- **Implemented a multilingual semantic search pipeline** using Cohere embed-multilingual-v3.0 and Qdrant vector database, enabling accurate Arabic and English document retrieval with configurable similarity thresholds.
- **Delivered a dual-interface system** — an interactive Streamlit UI for end users and a FastAPI REST API for programmatic integration — fully containerized with Docker Compose for one-command production deployment.

Arabic Handwritten Character Recognition — Multi-Stage Deep Learning Pipeline | *PyTorch · U-Net · EfficientNet-B1 · CNN · Triplet Loss · Grad-CAM* · [LINK](#)

- **Benchmarked CNN-from-scratch vs. EfficientNet-B1 transfer learning** on 16,800 Arabic handwritten character images (28 classes), achieving 98.72% test accuracy and 0.987 macro F1 — a +59% gain over the scratch model.
- **Built a full image retrieval system** using cosine similarity over learned embeddings, reaching 98.33% Top-1 and 99.17% Top 5 accuracy, then refined the embedding space with Triplet Loss metric learning and validated with t-SNE visualization.
- **Developed an end-to-end pipeline** with underwater-specific augmentations, Grad-CAM explainability heatmaps, and an interactive Gradio demo — all in a single self-contained Colab notebook.

YouHealth — AI Medical Chatbot 2nd Place, SoundAI Hackathon (SDAIA) | *LLM · lablab MENA* [LINK](#)

- Built AI assistant that converts patient conversations into structured medical reports.
- Nominated for GAIA AI Accelerator. Collaborated on full system design, NLP pipeline, and deployment.

PhishWarden — Phishing Detection System (Graduation Project) | *Python · Flask · ML · NLP · HTML/CSS/JS* [LINK](#)

- ML-based system detecting phishing emails and URLs via NLP feature extraction and classification.
- Full-stack implementation with Flask backend — key contributor to (A+) and GPA 4.95/5.0 First-Class Honors.

SKILLS

ML / DL Engineering	TensorFlow · scikit-learn · NLP · NLTK · Hugging Face · Deep Learning · Python · Git SQL · Flask · FastAPI · REST APIs · Linux · Power BI · Cloud Deployment AWS · Docker
Computer Vision	YOLO · U-Net · Semantic Segmentation · Object Detection · PyTorch
Agentic AI	LangGraph · LangChain Agents · CrewAI · AutoGen · OpenWebUI · n8n (workflows & custom nodes)
LLMs & RAG	RAG Pipelines · Prompt Engineering · OpenAI API · Ollama · ChromaDB · Embeddings
Soft Skills	Self-directed · Problem Solving · Leadership · Communication · Adaptability

CERTIFICATES & TRAINING COURSES

• Apple Developer Academy — Apple	Nov - 2025
• Machine Learning Specialization — Stanford	Dec - 2025
• IBM Machine Learning Professional Certificate — IBM	May - 2025
• Associate AI Engineer for Data Scientists — DataCamp	Jul - 2025
• Deep Learning Specialization — DeepLearning.AI	May - 2025
• AI Program (Introductory & Advanced) — KAUST	Feb - 2025
• Python 3 Programming — University of Michigan	Nov - 2024
• McKinsey Forward Program — McKinsey Company	Dec - 2024
• Track of Artificial Intelligence — FutureX	Mar - 2023